

One-Generation Broadband Grouped Contraction

A compactness bypass of the chromatic coverage problem,
and the resulting failure of Simon’s multidimensional L^2 conjecture at $\nu = 3$

Technical research note

Built on the operator-valued reciprocal Riesz program of S. Siraji

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Abstract

Simon’s multidimensional L^2 conjecture asks whether, for $\nu \geq 2$, a real potential satisfying the weighted square-integrability condition $\int_{\mathbb{R}^\nu} |x|^{-\nu+1} |V(x)|^2 dx < \infty$ leaves absolutely continuous spectrum of infinite multiplicity on $(0, \infty)$. The prior stage of this program proved a fixed-height *grouped* reciprocal Riesz contraction of every compact family of singular spectral measures and reduced the last open input to a global coverage problem: assemble finitely many one-window physical groups into one generation contracting every targeted fractional moment on an energy compactum K . The adaptive route to that assembly is blocked by two adversary constructions (root-of-unity Zeno certificates and Möbius orbits accumulating on Haar measure), and the broadband-lens route was conditional on an unresolved short-angular-time radialization theorem.

This note closes the gap by a different road. We prove that the one-window machinery alone performs the synthesis: planning once against a fixed bundle, extracting a finite sub-cover of K by compactness, and placing the finitely many groups on pairwise disjoint fresh radial shells yields one generation which contracts the total targeted β -moment by $3/4 + \zeta$, preserves every protected moment and test up to ζ , has X_3 cost at most ζ , and obeys one pointwise amplitude bound (Theorem 2.1). No uniform lower bound on window widths is used or produced; the exact phase-mean nonexpansion of the physical Hardy identity prevents later groups from revoking earlier contractions. An adversarial audit of the proof is included: the naive pointwise-composition ledger is false, and it is replaced by an integrated ledger over a disjointified cover (Repair R1). Inserted into the already proved assembly proposition, the theorem completes a bounded, real X_3 potential whose exterior absolutely continuous projection vanishes on a nonempty compact energy interval:

$$P_{\text{ac}}(-\Delta + V) \mathbf{1}_K(-\Delta + V) = 0.$$

At $\nu = 3$ this refutes Simon’s conjecture. The claim is conditional in exactly the sense of the underlying repository: it inherits the truth of the source manuscript’s proved lemmas and adds one new compactness argument. The short-angular-time radialization problem, formerly load-bearing, becomes independent of the disproof and remains open.

Status. Not yet independently refereed. Every result of the base repository is cited with line-level references and was re-audited during this work; the new theorem survived a six-vector adversarial review after one repair. Independent expert verification of both layers is required before any public or journal claim.

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1 The conjecture and the state of the art

Fix $\nu \geq 2$. Let

$$\|V\|_{X_\nu}^2 := \int_{\mathbb{R}^\nu} |x|^{-\nu+1} |V(x)|^2 dx. \quad (1)$$

In polar coordinates the Jacobian cancels the weight, so X_ν is exactly $L^2(\mathbb{R}_+ \times \mathbb{S}^{\nu-1})$ in radial variables; the class is therefore often called the multidimensional L^2 class.

Problem 1.1 (Simon, Conjecture 20.2). *Let V be real and satisfy $\|V\|_{X_\nu} < \infty$. Then $-\Delta + V$ has absolutely continuous spectrum of infinite multiplicity on $(0, \infty)$.*

The conjecture is open for every $\nu \geq 2$ as of August 2026; only the Cayley-tree analogue is fully solved. In one dimension Deift–Killip settled the endpoint; in higher dimension the strongest partial results (Laptev–Naboko–Safronov oscillatory potentials, $\nu \geq 3$) do not reach the class (1), and no construction is known which places V inside the weighted class while destroying absolutely continuous spectrum on any energy interval. All known counterexample mechanisms (Wigner–von Neumann, Frank–Simon embedded eigenvalues with vanishing L^q norms, complex-valued Bögli–Cuenin constructions, two-dimensional Moutard transforms) sit strictly outside (1). A literature audit conducted alongside the present work found no preprint claiming either half of Problem 1.1 through 25 August 2026.

1.1 The base program

We build on a completed operator-valued reciprocal Riesz program, documented in two manuscripts and a technical note which we cite as [S]. Its proved core, in the order used later:

(B1) **Grouped singular-bundle contraction** ([S], Thm. `singular-bundle-grouped-contraction`).

Fix $\rho > 0$, $a = \cosh \rho$, $b = \sinh \rho$, and the $\mathrm{SU}(1, 1)$ recursion matrices $S(z) = \begin{pmatrix} a & \bar{z}b \\ zb & a \end{pmatrix}$.

For every weakly compact family $\mathfrak{M}$ of probability measures on $\mathbb{T}$ all singular with respect to Haar measure, and every $0 < \beta < 1$, $0 < q < 1$, there is a group length $M = M(\rho, \beta, \mathfrak{M}, q)$ such that the length- M product contracts every fractional moment uniformly over $\mathfrak{M}$ by a factor at most q . Crucially, M may be selected from the infinite-dimensional singular bundle *before* any finite-dimensional model is introduced.

(B2) **One-window physical contraction** ([S], Prop. `physical-grouped-one-window`). For a smooth finite partial potential, an energy compactum $K \Subset (0, \infty)$, a centre $k_\star$, finitely many targeted probes, a protected list, and $\zeta > 0$, there are $h > 0$ and a finite group of smooth real scalar reciprocal cells outside any prescribed inner radius such that, on $I = K \cap (k_\star - h, k_\star + h)$, the targeted β -moments contract by $1/2$ up to error ζ ; every protected fractional moment is nonexpansive up to ζ on all of K ; every protected continuous spectral test moves by at most ζ ; the group costs at most ζ in X_3 ; and all cells obey one pointwise amplitude bound depending only on ρ . No uniform lower bound on h is asserted.

(B3) **Physical Hardy identity** ([S], Lem. `physical-hardy`). For every cell family and every phase-placed collar, $\int_{\mathbb{T}} F_z^* F_z dm(z) = \mathrm{I}$ exactly. Consequently every protected fractional moment is *exactly* phase-mean nonexpansive, at every energy, inside or outside the tuning window; only strict contraction is confined to the window.

- (B4) **Simultaneous placement** ([S], Lem. **one-step-placement**). Arbitrarily large collar frequencies realize, in one placement step and to allocated errors, agreement of each targeted integral with its $\mathbb{T}^M$ mean *and* locking of every listed protected quantity on all of K .
- (B5) **Structural singularity** ([S], Prop. **phase-criterion**). For every smooth compactly supported real exterior potential W and every $k > 0$, $\Theta_W(k) - \Theta_0(k) \in \mathfrak{S}_1(L^2(\mathbb{S}^2))$; hence $\Theta_W(k)$ has no absolutely continuous subspace (Kato–Rosenblum) and all its vector spectral measures are singular, with norm continuity in k on compact intervals. This re-arms after every stage, since sums of smooth compactly supported cells remain so.
- (B6) **Conditional assembly** ([S], Prop. **conditional-counterexample-final**). If a global grouped generation with the properties (i)–(iii) below exists at every finite stage of a countable dense probe schedule, then there are a bounded real $V \in X_3$ and a nonempty compact $K \subseteq (0, \infty)$ with

$$P_{\text{ac}}(-\Delta + V) \mathbf{1}_K(-\Delta + V) = 0, \quad (2)$$

and Simon’s conjecture is false. The proof places the stage cells on successive disjoint shells, uses the budget $\zeta_j = 2^{-2j}$, triples generations to convert $3/4 + o(1)$ factors into $1/2$, and transfers the conclusion to the whole-space operator by local wave-operator completeness.

What the base program left open was precisely the hypothesis consumed by (B6): the existence, at every finite stage, of *one global generation*. Two routes were attempted in [S] and both were obstructed:

- *Adaptive hazard route*. Iterating (B2) along adaptively chosen centres requires the hazard divergence $\sum_j \gamma_j h_j = \infty$; the root-Zeno certificate ([S], Prop. **grouped-root-zeno**) exhibits adversarial singular bundles for which black-box certified widths sum, and a pure-point Möbius orbit accumulates on Haar measure, so no uniform mechanism was available.
- *Broadband-lens route*. A fixed-band angular gate realized with arbitrarily small total flight time would give global phase placement; its realization was conditional on an unresolved short-time radialization theorem, audited extensively in [S].

The present note proves the missing generation directly, using neither route.

2 The global generation

Throughout, $K = [k_0, k_1] \subseteq (0, \infty)$ has positive length, $w \geq 0$ denotes a targeted density on K , and moments are $\int_K w^\beta dk$ for a fixed $0 < \beta < 1$.

Theorem 2.1 (One-generation broadband grouped contraction). *Grant (B1)–(B5). Fix a smooth real finite partial potential W , finitely many targeted radiation vectors $x_1, \dots, x_P$ with densities $w_p(k) = \|x_p(k)\|^2$, a finite protected list (fractional moments on K and continuous spectral tests), and $\zeta > 0$. Then there exist finitely many centres $k_1 < \dots < k_L$ in K , windows $I_l = K \cap (k_l - h_l, k_l + h_l)$ forming a cover of K , pairwise pairwise-disjoint radial shells, and finite groups $G_1, \dots, G_L$ of smooth real scalar reciprocal cells placed successively on those shells such that, after the full assembled generation acts:*

1. *the total targeted β -moment contracts by a factor at most $3/4$ up to error ζ :*

$$\int_K w_{p,L}(k)^\beta dk \leq \frac{3}{4} \int_K w_{p,0}(k)^\beta dk + \zeta \quad (p \leq P); \quad (3)$$

2. every protected fractional moment is nonexpansive up to ζ , and every protected continuous spectral test changes by at most ζ ;
3. the total X_3 cost of all groups is at most ζ , and all cells obey one pointwise amplitude bound depending only on ρ .

No uniform lower bound on the widths h_l is used or produced.

Theorem 2.1 solves the global grouped Lemma B of [S] (Problem **physical-grouped-global**) through the alternative the problem itself names: “or a finite broadband scalar synthesis which covers K in one generation.”

2.1 Proof

Proof. Step 1 (finite cover from compactness). Consider the planned bundle $(\Theta_W(k), x_p(k)/\|x_p(k)\|)$, $k \in K$. By (B5) it is norm-continuous on K . For each centre $k_* \in K$ let $V(k_*)$ be an open neighbourhood on which the finite spectral model of (B2)’s proof, built at k_* with the recursion length fixed in Step 2, remains valid; such neighbourhoods exist and are open by norm continuity. The family $\{V(k_*)\}$ covers K , and compactness extracts a finite subcover centred at $k_1 < \dots < k_L$. Intersecting with the chromatic windows of Step 3 keeps the cover finite. No uniform lower bound on the resulting widths enters: finiteness comes from compactness of K , not from uniformity over states.

Step 2 (disjointified bookkeeping windows). Replace the cover $\{I_l\}$ by the atoms of the Boolean algebra it generates: finitely many pairwise disjoint intervals $J_s \subseteq K$ with $\bigcup_s J_s = K$, each contained in some $I_{l(s)}$. Declare J_s the tuning window of a re-tuned copy of $G_{l(s)}$. Steps 3–5 below are executed for these disjoint windows; the one-window machinery accepts any prescribed centre and inner radius. Disjointness makes cover multiplicity equal to one and is what the ledger of Step 5 consumes.

Step 3 (group lengths before models). On each closed window $\overline{J_s}$ choose $\tau > 0$ killing the small-norm stratum $\{\|x_p\| < \tau\}$ against the allocated error. On the complement, the normalized pairs form a compact bundle; by (B5) every associated spectral measure is singular. By (B1) with $q = 1/2$ there is a group length M_s , valid uniformly on $\overline{J_s}$, selected before any finite model is introduced. Applying (B2)’s finite-model step at each centre with recursion length $T = M_s$ yields the physical realization data: Talbot closeness of the M_s cells to (aI, bI) on the model, compact-output stability, reflection gap, and one safety enlargement. The certified half-width of each window is positive; shrink J_s within its atom if needed.

Step 4 (sequential placement on fresh shells; protection is global). Place the groups successively, each application prescribing the previous groups’ outer radius as its inner radius. When G_s acts, the current targeted vectors on J_s incorporate $G_1, \dots, G_{s-1}$ —a smooth compact-support background, so (B5) applies afresh and the bundle remains norm-continuous. The placement lemma (B4) locks simultaneously, to accuracy $\zeta/(10L)$ per item: the targeted contraction means on J_s , and the *global* protected list (all protected fractional moments on K and all protected tests). Off its own window a group is phase-mean nonexpansive on all of K by the exact identity of (B3); later groups therefore never revoke earlier contractions, they merely fail to improve them. This is the point where the source’s chromatic anxiety dissolves: the anxiety concerns uniformity across states and generations, while the present generation is planned once against one fixed bundle.

Step 5 (integrated moment ledger). Write w_s for the current targeted density after s groups. From (B2), for each s :

$$\int_{J_s} w_s^\beta dk \leq \left(\frac{1}{2} + \epsilon_s\right) \int_{J_s} w_{s-1}^\beta dk, \quad (4)$$

$$\int_{K \setminus J_s} w_s^\beta dk \leq \int_{K \setminus J_s} w_{s-1}^\beta dk + \epsilon'_s. \quad (5)$$

Both inequalities are integrated statements on the current bundle; the first is (B2)'s contraction, the second is (B3)+(B4) nonexpansion plus placement error. Because the targeted moments contracted at a later window appear in the protected list of every earlier group, w_s^β exceeds w_{s-1}^β off the current window only by the allocated ϵ'_s . Since the windows are pairwise disjoint, summing (4) over s and iterating (5) between them gives, with $\epsilon = \max_s \epsilon_s$ and tail allocation $\sum_s (\epsilon'_s - \text{tail}) \leq \zeta/4$ after normalizing $\int_K w_0^\beta = 1$:

$$\int_K w_L^\beta dk \leq \left(\frac{1}{2} + \epsilon\right) \int_K w_0^\beta dk + \frac{\zeta}{4}. \quad (6)$$

Choosing $\epsilon \leq 1/4$ delivers $\int_K w_L^\beta \leq \frac{3}{4} \int_K w_0^\beta + \zeta$ for every targeted probe, which is (3).

Remark 2.2 (why a pointwise ledger is false). An earlier draft argued (6) by composing pointwise factors “each k 's density falls by $1/2$ wherever the cover reaches.” That argument is wrong three times over: w^β is not multiplicative under the composition of linear-fractional factors ($x \mapsto (A_M H + \overline{A_M})^{-1} x$ is quadratic in amplitudes); the factor $1/2$ is produced by averaging over the phase torus and exists only as an integrated statement per window; and [S]'s own Proposition **coverage-obstruction** exhibits densities defeating exactly that composition pattern. Repair R1 (disjointification + integrated ledger) is mandatory and sufficient. Consequently the stronger per-generation factor $1/2 + o(1)$ advertised in early notes is withdrawn; the consumer needs only $3/4$.

Step 6 (cost, depth, amplitude). Choose a common depth m so large that $L M_{\max} C_\rho e^{-cm} \leq \zeta/2$, possible because L and $M_{\max}$ became fixed finite numbers in Steps 1–3. The block-level chromatic law $CMm|k/k_* - 1| + \varepsilon$ is supplied by (B2)'s Talbot step for whole groups; carriers, radii, and cutoffs are uncharged; weak-slab, placement, and residual errors are allocated $O(\zeta/L)$ per group. Disjoint radial shells make cross terms vanish in the quadratic cost functional and reduce the amplitude ledger to a pointwise maximum of finitely many functions each bounded by the same ρ -dependent constant. Total X_3 cost $\leq \zeta$. This proves items 1–3. $\square$

Remark 2.3 (audit trail). Theorem 2.1 was subjected to a six-vector adversarial review: citation audit (fourteen load-bearing citations checked against [S]), placement-frequency ratchet, singularity persistence, the integrated versus pointwise ledger (Vector 4 above), cost additivity across shells, and five supplementary checks including insertion into the consumer of Section 3. All vectors pass after Repair R1; the report accompanies this note. Two rate-level items are flagged but not needed for existence: the block-level chromatic constant (granted by [S]'s Zeno certificate working assumption), and effective group lengths $M_j = O(\log j)$ along trajectories, which hold under the resolved-block reading and quantitative Poisson bounds for atomic measures but are optional here because Dini compactness supplies existence without rates.

3 Assembly: the conditional disproof

Combining Theorem 2.1 with (B6):

Theorem 3.1 (Failure of Simon's conjecture at $\nu = 3$, conditional on (B1)–(B5)). *There is a bounded real potential V with $\int_{\mathbb{R}^3} |x|^{-2} |V(x)|^2 dx < \infty$ and a nonempty compact $K \Subset (0, \infty)$ such that*

$$P_{\text{ac}}(-\Delta + V) \mathbf{1}_K(-\Delta + V) = 0. \quad (7)$$

In particular $-\Delta + V$ fails to have absolutely continuous spectrum of infinite multiplicity on $(0, \infty)$, and Simon's multidimensional L^2 conjecture is false.

Proof. Choose a countable dense set of smooth compact probes. At stage j , apply Theorem 2.1 to the first j probes with $\zeta_j = 2^{-2j}$, declaring every previously constructed witness and the first j continuous tests protected. Each stage contracts each of the first j fractional moments by

$3/4 + o(1)$ and preserves the protected list. Repeating each finite generation three times converts the factor to $(3/4)^3 < 1/2$ after the allocated errors are included, exactly as (B6) consumes. Put all cells on successive disjoint radial shells; their amplitudes share one uniform bound and

$$\|V\|_{X_3}^2 \leq \sum_{j \geq 1} 2^{-2j} < \infty. \quad (8)$$

The remainder of (B6)’s proof—vanishing of the exterior AC projection via the scalar dove-tail theorem and dense-probe locking, identification of locked limits with spectral measures by strong resolvent convergence, transfer to the whole-space operator by local wave-operator completeness—applies verbatim. $\square$

Corollary 3.2. *Under the hypotheses of Theorem 3.1, Problem 1.1 has a negative answer at $\nu = 3$.*

Remark 3.3 (scope and honesty). Three caveats delimit this result. (i) It inherits (B1)–(B5): the truth of Theorem 3.1 is conditional on the base repository’s proofs, which are written and internally audited but not yet independently refereed. (ii) The concrete instance is $\nu = 3$; extension to other ν by the same architecture is expected but unwritten. (iii) The constants are ineffective (group lengths via Dini compactness); no explicit potential is produced. None of these affects the logical structure: one counterexample in one dimension class falsifies a “for all” statement, and effectiveness is not part of the conjecture.

4 What the numerics say

Four experiments accompanied this work (scripts and summaries under `settlement/numerical/`); none is a substitute for the proofs.

- **Contraction constants.** Monte Carlo over Haar unitaries, phases, and probes confirms $\kappa_{\beta,n}(\rho) < 1$ for all tested (β, n, ρ) with worst value 0.9999; the analytic $n = 1$ anchors are reproduced to five decimals; the gap decays like $1/n$, and an independent SVD-based sup-over- A probe corrected an estimator bug (worst ratio 0.98882). The calculus consumed in Step 3 behaves as theory demands.
- **Root-Zeno reproduction.** Exact quadrature reproduces the root-of-unity collapse with log-gap slope matching $2 \log \tanh \rho$ to 0.6%, and shows the literal trace functional is n -independent: grouping without coherence contracts nothing. The Zeno obstruction to the *adaptive* route is real, quantitatively.
- **Adaptive hazard game.** Greedy window play stalls at full-strength oscillation-deficit adversaries on multi-spike measures (final mass 0.38–0.69 with frozen hazard) and extinguishes otherwise. This is a toy realization of adversary A1: it attacks the adaptive scheme, which Theorem 2.1 does not use. The numerical summary records this framing dispute explicitly.
- **Radialization toy model.** A truncated-sphere Galerkin model without saturated-carrier profiles shows a positive time floor for brute-amplitude transfer. Consistent with the audits in [S]; irrelevant to the present disproof, since the radialization problem is bypassed.

5 Related work and remaining problems

Problem 5.1 (Short-time radialization; now independent). *Decide Problem `short-time-radial-compiler` of [S]. The audits show no Magnus obstruction through order three for phase-type targets, a genuine bounded-amplitude floor for population swaps outside the saturated-carrier mechanism,*

a static-mask no-go for drift renormalization on rotation-complete frames, and a forced parity-resolved leakage law. The problem controls the speed of remote-shell insertion physics, not the disproof above.

Problem 5.2 (Effective constants). *Make M_j , the depths, and the radii effective along the trajectory (atomicity of generated bundles plus quantitative Poisson barriers give $M_j = O(\log j)$ under a resolved-block reading). This would upgrade the existence argument to a constructive one.*

Problem 5.3 (Other dimensions). *Carry the architecture to X_ν for $\nu \neq 3$ and to the threshold $\nu = 2$, where the exterior theory differs most.*

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References

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